Guide to the lab

The Enzyme's Handshake · BCH 3023 · Session 1, baker's yeast reduces a ketone; Session 2, a lipase sorts two enantiomers

In two sessions, enzymes do organic chemistry and you measure how well they tell mirror-image molecules apart. In Session 1, baker's yeast reduces ethyl acetoacetate to ethyl 3-hydroxybutanoate, mostly one enantiomer; you work it up, weigh it, measure its rotation and run its NMR. In Session 2, the lipase CAL-B acylates one enantiomer of a racemic alcohol faster than the other; you follow it on a chiral GC, stop it near halfway, separate the two products and work out how selective the enzyme was. This guide takes you through every step: **what to do**, **what you should see**, and **why**. Read the prelab review first (linked under the lab page's title).

Before you start

- **Your code** is on the page (Session 1, beside your recipe; Session 2, beside your alcohol). It sets how your yeast is fed in Session 1 and which alcohol you resolve, with how much enzyme, in Session 2. Use the same code, computer and browser for both: the page keeps your work there.
- Session 1 is the lab page as it opens; Session 2 is the same page with **?part=2** at the end of its address (the link is in the note under the title).
- Every step can be done in the 3D room, by clicking the thing it uses (a gold outline shows what a click will use, and the line under the room says what it does), or with the step's button in the list. The next step to do is marked in the list. This guide names the buttons.
- Days of fermentation and hours of shaking pass on the **bench clock**: each wait button moves it on.
- The pictures here were taken on one code; your recipe, your numbers and, in Session 2, your alcohol will differ.

In a real lab

- **Hexane, ethyl acetate and vinyl acetate** are highly flammable: no flames, and vinyl acetate in the hood. **Chloroform** (for the polarimeter) is toxic: make up solutions in the hood, with gloves.
- **Celite and silica gel** are fine dusts that harm the lungs: keep them wet, and handle silica in the hood.
- **Novozym 435** (the enzyme beads) is a respiratory sensitizer: weigh it gently, without dust.
- **Vacuum glassware** can implode: clamp the filter flask, put a trap before the pump, and check the rotary evaporator's flasks for cracks.

SESSION 1

The yeast

1 Wake the yeast

Open The yeast view: the water bath, the shaker incubator, the stirring hotplate with the 1 L flask, and the reagent jars. Set the water bath (30 °C), pour in the warm water (Pour, or click the flask), then weigh and add the sucrose and the baker's yeast (Add, or click each jar). Your recipe, shown at the top of the section, says how much and whether your yeast was starved in 5% ethanol instead.



The yeast's corner, a live culture foaming in the flask on the hotplate.

You should see the broth foam: the yeast is alive and fermenting the sugar.

If the water is too hot (60 °C), the page warns you: the yeast dies, does not foam, and reduces nothing.

Why sugar? The cells burn it to keep making the NADPH their reductases use.

2 Add the ketone, and let the days pass

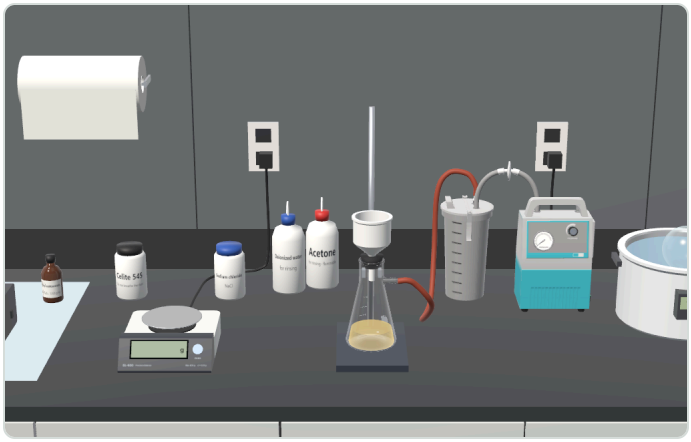
Add the ethyl acetoacetate as your recipe says (Add beside each portion, or click the brown bottle), with a second feed of sugar where your recipe has one. Move the clock on with 1 hour, 6 hours or A day (or click the bench timer for 6 hours). When the clock reaches your recipe's time, run a TLC of the broth (Run it , or click the TLC jar in the hood).

You should see the TLC's result under the step list: the blue spot of ethyl acetoacetate while it is still there, then none.

Stopping while the spot is there leaves ketone in your product: a lower yield, a rotation that reads low, and a peak in your NMR.

3 Filter off the yeast through Celite

Build the Celite bed in the Büchner funnel (Build it , or click the Celite jar), stir Celite into the broth (Stir in), and filter by suction (Filter , or click the Büchner funnel).



The filtration: the Büchner funnel on its clamped filter flask, the red vacuum hose to the trap and the pump. The filtrate is in the flask.

Without the bed, the yeast clogs the paper and the filtration crawls.

4 Salt the filtrate and extract

Saturate the filtrate with salt (Add salt, or click the sodium chloride jar), then extract it three times with 50 mL of ethyl acetate, gently (Gently, or click the separatory funnel in the hood).



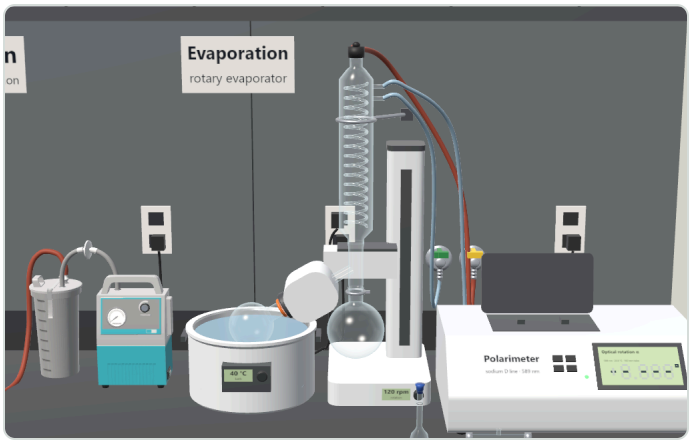
In the hood: the separatory funnel with its two layers, the water layer below and ethyl acetate above.

Shaken hard (Shake hard), the yeast debris makes an emulsion. Clear it with Brine, Wait or Centrifuge before you go on.

Why salt? Ethyl 3-hydroxybutanoate likes water; salt pushes it into the ethyl acetate.

5 Wash, dry, filter and evaporate

Wash the ethyl acetate with brine (Wash), dry it over magnesium sulfate in portions until it no longer clumps (Add a portion), gravity filter it into the weighed round-bottomed flask (Filter), and take off the solvent on the rotary evaporator (Evaporate; in the room, the rotary evaporator does the filtering and then the evaporating). Then weigh the flask (Weigh, or click the balance).



The rotary evaporator: the flask turning in the 40 °C bath, the condenser cooled from the green tap, the vacuum from the yellow one.

Droplets left mean the solution was not dry: Redissolve and dry again.

You should see the balance's two readings, the flask empty and with your product. Work out your percent yield from them.

6 Measure the rotation

At the polarimeter (Polarimeter view): click the 25 mL volumetric flask (or Weigh a sample, make up the solution): about 0.25 g of your product, made up to the mark with chloroform in the hood. Click the tube on its stand (or Zero on a blank): it is filled with chloroform, laid in the instrument, and the zero is set. Click the tube again (or Fill the tube carefully): it is rinsed and filled with your solution. Click the instrument (or Read the angle).



The digital polarimeter with the tube in its compartment; the display reads α.

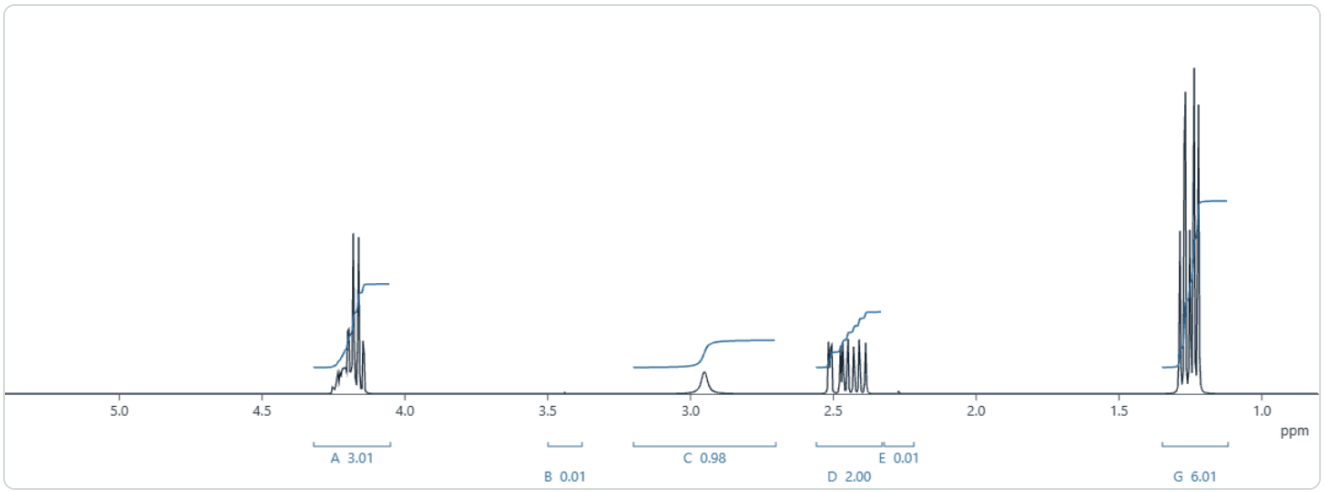
You should see dashes until the zero is set, then 0.000, then your angle. The page also writes the angle and your sample's mass under the buttons.

Mistakes: Fill it quickly leaves an air bubble and the angle reads low; reading without the zero leaves the instrument's offset in it.

To answer: [α] = α / (l c), with l in dm and c in g per mL; the ee is your [α] as a share of +43.5°, pure (S)-ethyl 3-hydroxybutanoate's.

7 Run the NMR

Click the NMR sample table (NMR sample table view; or Make the NMR sample), then the magnet (or Run the ^{1}H spectrum).



A product's spectrum, 0.8 to 5.4 ppm, with its integral traces and regions. Point at the spectrum to read a shift; the buttons above it show the whole range and a close-up of 2.2 to 2.6 ppm.

To answer: assign each signal from its shift, its splitting and its integral; compare region B (the ketone's CH_2, 2 H) with region D (the alcohol's CH_2, 2 H) for the mole percent of ketone left; then correct your ee for the ketone that weighed into the polarimeter sample without rotating.

SESSION 2

The lipase

8 Set up the resolution

Weigh 1.00 mmol of your racemic alcohol into a 20 mL vial (Weigh , or click the balance), add 5 mL of dry hexane (Dry hexane) and 2.5 equivalents of vinyl acetate (2.5 equiv), set the shaker incubator (40 °C , or click it), and add your handout's Novozym 435 beads (Add … mg , or click the fridge). The bench clock starts.

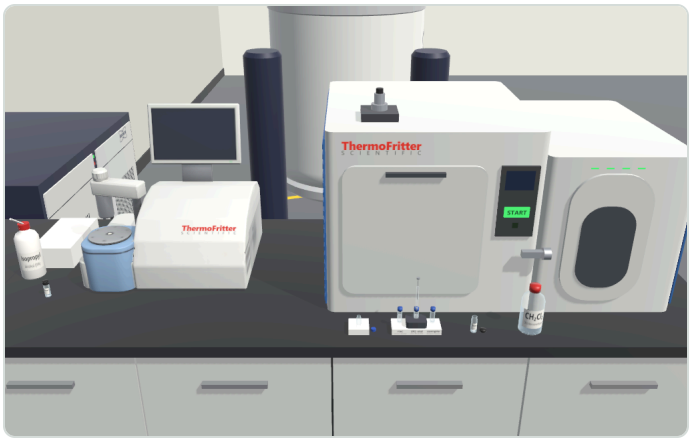


The vial on the shaker's platform, with its hexane and the enzyme beads.

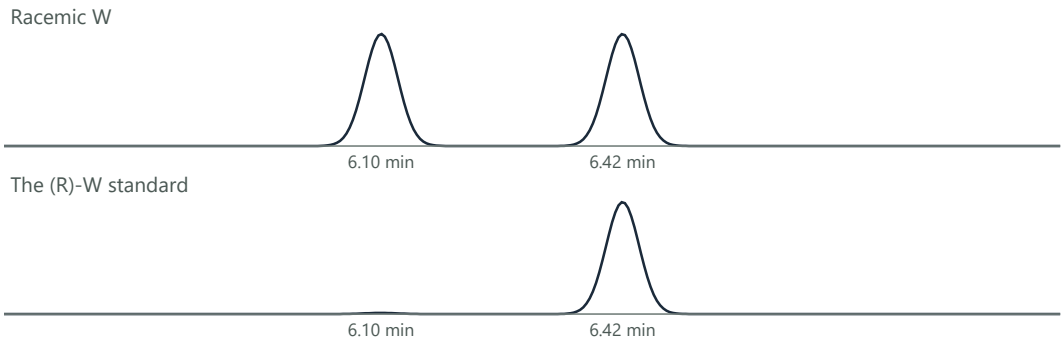
> **Mistakes:** Hexane from the wash bottle is wet (the lipase hydrolyses too); 1 equiv runs out; 60 °C is faster but less selective; three times the beads is faster and, for some alcohols, less selective.

9 Run the standards on the chiral GC

Run your racemic alcohol (Racemic alcohol) and the pure (R) standard ((R) standard), or click the GC (IR and GC-MS view; before the enzyme is in, a click runs the racemic alcohol). Each run is drawn under the step list, newest first, with the integrator's report: each peak's number, time, area and area percent.



The GC, here fitted with a chiral cyclodextrin column.



A made-up example (compound W, not one of the lab's): the racemic sample (top) gives two peaks of equal area; the pure (R) standard (bottom) comes off with the second, so the second peak is (R).

> **Why a standard?** The order of the two enantiomers on a chiral column is not a rule: it changes from one compound to the next. The standard's time names your peaks.

10 Follow it, and stop near 50%

Let it shake (15 min , 1 h , 4 h , or click the shaker for 15 minutes) and take samples (Sample , or click the GC). Each sample's chromatogram shows the alcohol's two peaks and, growing as the reaction runs, its acetate's (and vinyl acetate and acetaldehyde near the start). When about half the alcohol has become acetate, stop it (Stop): the beads are filtered off.

> **You should see** one alcohol peak shrink while the other stays, and one acetate peak grow far more than the other. How fast depends on your alcohol: some reach half-way in under an hour, some take a day.

> **Mistakes:** Without the filter lets beads into the GC vial (it goes on reacting); Undiluted overloads the column, and close peaks run together into one. Going far past 50% makes the alcohol purer but the acetate less pure.

11 Separate the acetate from the alcohol

Try eluents on TLC (Hexane , 9:1 , 4:1 , 1:1 hexanes : ethyl acetate; the R_f values appear under the step list), then run the flash column in the one you choose (or click the column in the hood). Take off the solvent (Warm bath (40 °C) , or Cold bath, gentle for a volatile alcohol), weigh both fractions (Weigh), and run each on the GC (Acetate fraction , Alcohol fraction).



In the hood: the flash column with its rack of tubes, the TLC jar, the solvents.

> **Mistakes:** 1 : 1 runs both compounds off together, so each fraction holds some of the other; hexane alone leaves the alcohol on the column; a warm bath loses part of a volatile alcohol.

12 Work out what the enzyme did

From the alcohol fraction's two alcohol peaks, its ee (ee_s); from the acetate fraction's two acetate peaks, the acetate's (ee_p). Then the conversion, E, and the two yields from the balance's masses. The page checks each against your own numbers. Last, say which enantiomer the lipase acylated faster (from the (R) standard and which peak shrank) and whether the Kazlauskas rule predicts it: your instructor marks those two.

$c = ee_s / (ee_s + ee_p)$ · $E = \ln[(1 - c)(1 - ee_s)] / \ln[(1 - c)(1 + ee_s)]$, with c and ee_s as fractions · $ee = |A_1 - A_2| / (A_1 + A_2)$

13 Print your report

At the end of each session, type your name under **Your report** and press Print or save as PDF; choose "Save as PDF" to hand it in as a file. It prints your results, spectrum or chromatograms and a summary for your instructor: what the page accepted, what the bench warned you about, and (Session 2) your two answers for marking.

REFERENCE

The bench's warnings, and what to do

The page says	What it means	What to do
That water is hot enough to kill yeast…	Above about 55 °C the yeast dies.	Start over with the bath at 30 °C.
The TLC still shows ethyl acetoacetate…	The reduction is not finished.	Let more time pass and run the TLC again.
Without a Celite bed…	The paper clogs.	Build the bed before filtering.
Shaken hard… an emulsion	No sharp line between the layers.	Brine, waiting or the centrifuge.
The magnesium sulfate is still clumping	The solution is still wet.	Add another portion.
Droplets in the residue	It was not dry when evaporated.	Redissolve and dry again.
An air bubble sits in the light path	The angle reads low.	Fill the tube carefully and read again.
That hexane was not dried	The lipase hydrolyses as well as acylates.	Start Part 2 over with dry hexane.
Warm… tells the enantiomers apart less well	E falls with temperature.	40 °C is the usual compromise.
An undiluted sample…	Overloaded peaks run together.	Take a diluted sample.
Beads went into the GC vial	The sample kept reacting.	Sample through the pipette filter.
Too polar… / Hexane alone…	The column will not separate them / the alcohol stays on it.	Choose the eluent by TLC first.

A simulation for BCH 3023. The pictures were taken on one code; the chromatogram in step 9 is made up and is of no compound in the lab.